



RTL Design Sherpa

Reed-Solomon Board Validation Report Genesys 2 FPGA board evidence

October 2026

Table of Contents

1 Document Information.....	8
1.1 Revision History.....	8
1.2 Scope.....	9
1.3 Provenance.....	10
2 Scope and Claims.....	11
2.1 What this book is.....	11
2.2 What is claimed.....	12
2.3 What is not claimed.....	13
3 Images and Codec Profile.....	14
3.1 Image matrix.....	14
3.2 Codec profile.....	15
4 Harness Architecture.....	16
4.1 RTL inventory.....	16
4.2 Host and build inventory.....	18
4.2.1 Figure 2.1: Reed-Solomon AXIS datapath and register path.....	18
4.3 Register path, clock, and reset.....	19
5 Campaigns and the Oracle Argument.....	20
5.1 Campaign inventory.....	20
5.2 Determinism and cross-solver agreement.....	21
5.3 The dual-solver loop as oracle.....	22
6 Testing Choices.....	23
6.1 Injector modes mapped to failure mechanisms.....	23

6.2 Post-encoder injection protects parity symbols.....	24
6.3 UART and the safety pair.....	25
6.4 Genesys 2 vs Nexys A7.....	26
6.5 AXIS and AXI4 flavors cover both integration styles.....	27
6.6 Deliberately not board-tested.....	28
7 Erasure Campaign.....	29
7.1 What erasure decoding adds.....	29
7.2 Board findings.....	30
7.3 TASK-005 boundary.....	31
8 Battery Results.....	32
8.1 Per-image campaign status.....	32
8.2 Correction boundary.....	33
8.2.1 Figure 4.1: Reed-Solomon correction boundary across all four images.	33
8.3 Decode cost.....	34
8.3.1 Figure 4.2: Reed-Solomon decode cost across all four images.....	34
8.4 Sweep verdict table.....	35
8.5 Solver agreement.....	37
8.5.1 Figure 4.3: Reed-Solomon solver agreement matrix (riBM vs Euclid)....	37
9 Soak Run Scheduled.....	38
9.1 Status at publication time.....	38
9.2 Target.....	39
9.3 What will be reported.....	40
10 Honest Limits.....	41

10.1 What the board evidence does not cover.....	41
10.2 Traceability.....	42

List of Figures

Figure 2.1: Reed-Solomon AXIS datapath and register path.....	18
Figure 4.1: Reed-Solomon correction boundary across all four images.....	33
Figure 4.2: Reed-Solomon decode cost across all four images.....	34
Figure 4.3: Reed-Solomon solver agreement matrix (riBM vs Euclid).....	37

List of Tables

Table 1: Table 0.1: Revision history.....	8
Table 2: Table 1.1: Reed-Solomon Genesys 2 image matrix.....	14
Table 3: Table 2.1: RTL blocks in the Reed-Solomon harness.....	16
Table 4: Table 2.2: Host scripts and build flow.....	18
Table 5: Table 3.1: Reed-Solomon board campaign inventory.....	20
Table 6: Table 3.2: Reed-Solomon erasure boundary results.....	30
Table 7: Table 4.1: Reed-Solomon 2026-10-05 battery per-image status.....	32
Table 8: Table 4.2: Reed-Solomon sweep verdicts.....	35
Table 9: Table 5.1: Traceability from claim to artifact.....	42

1 Document Information

1.1 Revision History

Table 0.1: Revision history.

Revision	Date	Author	Description
0.1	2026-10-05	RTL Design Sherpa	Initial board validation roll-up for the Reed- Solomon Genesys 2 harness.

1.2 Scope

This book covers the RS(252,236) $t=8$ codec as exercised on the Digilent Genesys 2 board. It describes the harness, the campaign methodology, the 2026-10-05 battery, the erasure boundary finding, and the scheduled million-block soak.

1.3 Provenance

All timing and campaign results in this book come from `stable/MANIFEST.md` and `stable/results/2026-10-05_battery/`. The board used for the runs is Digilent Genesys 2 serial 200300B818A0 (Kintex-7 xc7k325t-2), programmed and measured with Vivado 2025.1.

2 Scope and Claims

2.1 What this book is

This report is the board's statement about the RS(252,236) $t=8$ codec. The Genesys 2 harness runs the same campaigns in cocotb simulation and on real silicon, so when the board passes we have evidence that simulation matches hardware at 100 MHz with real I/O, real clocks, and real register traffic.

2.2 What is claimed

The 2026-10-05 battery demonstrates that four single-decoder bitstreams (two flavours times two solvers) pass the same deterministic campaign set:

- `init`, `smoke`, `sweep`, `random`, `clusters`, `localized`, and `badblock` all PASS on `axis_ribm`, `axis_euclid`, `axi4_ribm`, and `axi4_euclid`.
- The sweep shows $e \leq 8$ corrected, $e = 9 \dots 18$ uncorrectable, on every image.
- The random campaign recorded 64 runs x 16 blocks with 0 failing runs across all images.
- The matching verdict counts between riBM and Euclid images are themselves a cross-solver check.

That is board evidence that both key-equation solvers behave identically under the same workload and that the codec honours the correction boundary.

2.3 What is not claimed

This book does not replace the component DV matrix. It does not prove every possible error pattern or every smaller RS profile. The dual-decoder on-chip comparator is present in simulation but not in the production board matrix; agreement is proven by running identical deterministic campaigns on the two single-decoder images. The erasure $f = 2t$ boundary is a known open issue (TASK-005) and is called out in Chapter 3 and Chapter 5. The scheduled soak has not completed at publication time; Chapter 4 says what it will report.

3 Images and Codec Profile

3.1 Image matrix

Four Genesys 2 bitstreams were built and kept in `stable/reports/`:

Table 1.1: Reed-Solomon Genesys 2 image matrix.

Image	Solver	Bitstream	WNS (ns)	Failing endpoints	Notes
axis_ri bm	riBM	rs_loop_ge nesys2_axi s_ribm.bit	+1.072	0	Stream ing pipe.
axis_e uclid	Euclid	rs_loop_ge nesys2_axi s_euclid.b it	+0.875	0	Stream ing pipe.
axi4_ri bm	riBM	rs_loop_ge nesys2_axi 4_ribm.bit	+1.822	0	Memor y-to- memor y chain.
axi4_e uclid	Euclid	rs_loop_ge nesys2_axi 4_euclid.b it	+1.431	0	Memor y-to- memor y chain.

All four images are timing-clean at 100 MHz on the k325t-2. The WNS values were read from the four `stable/reports/genesys2_*/timing_summary.txt` files.

3.2 Codec profile

The geometry is frozen in `build-loop/rtl/rs_loop_cfg_pkg.sv`:

- Code: RS(252,236), $t=8$, $m=8$.
- Symbols per beat: 4 ($S=4$), so a 32-bit beat carries four 8-bit symbols.
- Shortened from the natural 255-symbol RS block so that $n=252$ and $k=236$ are both multiples of the beat size. A full codeword is 63 beats; the message is 59 beats.

The shortening was deliberate: it makes the AXIS datapath line-rate at 63 cycles per block and keeps the AXI4 memory alignment clean. The host reads `PROFILE` to discover the geometry and `TOPOLOGY` to discover which datapath and solver are loaded.

4 Harness Architecture

4.1 RTL inventory

Table 2.1: RTL blocks in the Reed-Solomon harness.

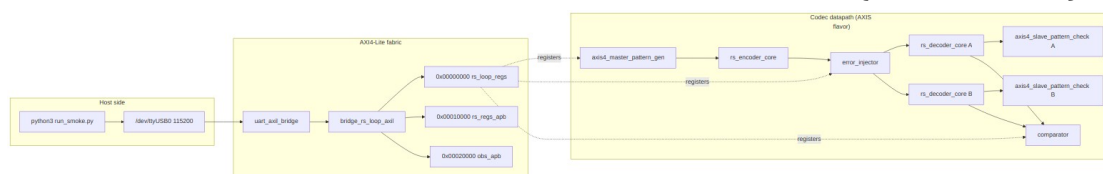
Module	File	Role
uart_axil_bridge	projects/components/ utility-ip/ converters/rtl/ uart_to_axil4/ uart_axil_bridge.sv	UART 115200 8N1 to AXI4-Lite master.
bridge_rs_loop_axil	Generated fabric	1-master x 3-slave AXI4- Lite/APB fabric.
rs_loop_genesys2_top / rs_loop_top	build-loop/rtl/ rs_loop_genesys2_top. sv / build-loop/rtl/rs_loo p_top.sv	Genesys 2 MMCM wrapper, or Nexys A7 100 MHz direct clock.
rs_loop_harness	build-loop/rtl/ rs_loop_harness.sv	Codec loop, register decode, bandwidth meters, observers, verdict tallies.
rs_loop_cfg_pkg	build-loop/rtl/ rs_loop_cfg_pkg.sv	Single source of geometry: RS(252,236), 4 symbols/beat.
rs_loop_regs	Generated from build- loop/rtl/rs_loop_regs .rdl	Host-visible CSR block.
axis4_master_pattern_ gen	rtl/amba/shared/ axis4_master_pattern_ gen.sv	LFSR data source plus expected CRC-32, one packet per block.
axis4_slave_pattern_c heck	rtl/amba/shared/ axis4_slave_pattern_c heck.sv	Regenerates the same LFSR pattern and compares beats per decoder.
rs_encoder_core	Component RS RTL	RS(252,236) encoder.
error_injector	Shared utility RTL	Post-encoder corruption:

Module	File	Role
		modes 0..7 plus erasure mark.
rs_decoder_core	Component RS RTL	Decoder with selectable KES_ALGO and erasure path.
rs_erasure_unit	Component RS RTL	Erasure locator / transform used when ERASURE_SUPPORT=1.
rs_axi4_pipeline	build-loop/rtl/rs_axi4_pipeline.sv	Memory-to-memory job chain for the AXI4 flavor.
axis4_intf_observer	Shared utility RTL	Four AXIS seam observer on window 0x20000.
axi4_intf_master_observer	Shared utility RTL	Four codec AXI4 master-port observer on window 0x10000.

4.2 Host and build inventory

Table 2.2: Host scripts and build flow.

Script / target	File	Role
rs_env.py	bin/rs_env.py	Path anchor for shared host code.
run_smoke.py	bin/run_smoke.py	Campaign runner.
init / smoke / sweep / random / soak / erasure	bin/seq_*.py	Deterministic campaign sequences.
RsLoopDriver	build-loop/host/rs_loop.py	By-name register access and status collection.
rs_loop_programs	build-loop/host/rs_loop_programs.py	Shared authored-once programs.
host_rs_loop.py	build-loop/host/host_rs_loop.py	CLI front-end.
make -C build-loop bitstream	build-loop/Makefile	Vivado build; set RS_TARGET=genesys2.
bin/build_image_matrix.sh	bin/build_image_matrix.sh	Builds {axis,axi4} x {riBM, Euclid}.



Datapath diagram

4.2.1 Figure 2.1: Reed-Solomon AXIS datapath and register path.

4.3 Register path, clock, and reset

The host register path is Python over pyserial to the UART, into `uart_axil_bridge`, through the generated `bridge_rs_loop_axil` 1x3 fabric, and out as three APB windows. Window 0 at `0x00000000` holds `rs_loop_regs` (`BUILD_ID`, `SCRATCH`, `PROFILE`, `TOPOLOGY`, `INJ_CFG`, etc.). Window 1 at `0x00010000` hosts the AXI4 master observer on AXI4 flavours and is a read-zero stub on AXIS. Window 2 at `0x00020000` is always live and hosts the AXIS seam observer.

On the Genesys 2 the 200 MHz LVDS system clock passes through IBUFDS into an `MMCME2_BASE` with `VCO = 1200 MHz` and `CLKOUT0_DIVIDE_F = 12`, producing the 100 MHz harness clock. The Nexys A7 top uses the on-board 100 MHz clock directly.

5 Campaigns and the Oracle Argument

5.1 Campaign inventory

Table 3.1: Reed-Solomon board campaign inventory.

Campaign	What it proves	Determinism
init	BUILD_ID, SCRATCH round-trip, PROFILE, TOPOLOGY.	Deterministic.
smoke	Bypass, clean, $e=t$, $e=t+1$, deterministic DEBUG walk.	Deterministic.
sweep	Exact correction counts for $e = 0 \dots 2t+2$.	Deterministic.
random	Fresh seeds, mixed injector modes, 64 runs x 16 blocks.	Host-seeded; reproducible on demand.
clusters	Spatially correlated errors.	Deterministic combo matrix.
localized	Windowed errors.	Deterministic combo matrix.
badblock	Retention-tail / worn-block behaviour.	Deterministic combo matrix.
erasure	Marked-symbol runs at $f = t, 2t, 2t+1$.	Deterministic.
soak	Long-running mixed-mode stress; the million-block run.	Host-seeded; replayable.

5.2 Determinism and cross-solver agreement

The host-side RNG seed defaults to 1. With that default, the random, soak, clusters, localized, and badblock campaigns are fully deterministic. Because the riBM and Euclid images run bit-identical workloads, the matching verdict counts between them are the cross-check. You do not need the on-chip comparator to know the two solvers agree; the campaigns prove it.

5.3 The dual-solver loop as oracle

The AXIS datapath in simulation feeds the same corrupted blocks to two `rs_decoder_core` instances, one with `KES_ALGO=RIBM` and one with `KES_ALGO=EUCLID`. Each decoder's output is compared beat-by-beat against the generator's regenerated LFSR pattern in its own `axis4_slave_pattern_check`, so a correction is judged by a reference that never saw the errors. A separate comparator then requires the two decoders to agree on every beat and on every block's verdict. On the board images this comparator is absent by construction; the cross-check is the deterministic campaign battery.

6 Testing Choices

6.1 Injector modes mapped to failure mechanisms

The shared `error_injector` operates on 8-bit symbols. Mode 1 exact-count places exactly e errors per block at uniformly random distinct positions, which is what makes the $e = t / e = t + 1$ boundary sharp. Mode 2 burst and mode 3 rate model raw channel bit-error behaviour. Modes 4 clusters and 5 localized model spatially correlated faults. Mode 6 badblock models the tail of NAND retention. Mode 7 debug is the deterministic bring-up walk.

6.2 Post-encoder injection protects parity symbols

Errors are injected after the encoder, not in the generator. An error in the generator's data would be encoded faithfully and would be invisible to the code. Placing the injector on the coded stream means parity symbols are corrupted too, which is the workload the decoder actually has to handle.

6.3 UART and the safety pair

The harness uses a plain UART instead of a proprietary debug core, so the host side is just pyserial and Python scripts. Board safety is a pair: `board_lock.sh` serializes programming flows by JTAG serial, and `fpga_board.py` reads the JTAG chain before programming to confirm the expected Genesys 2 serial (200300B818A0) is present with a device behind it.

6.4 Genesys 2 vs Nexys A7

The Kintex-7 XC7K325T-2 was chosen because it has room for the full harness plus observers and still closes timing with positive slack. The Nexys A7-100T flow is kept as a target option (RS_TARGET=nexys_a7_100t) but the Genesys 2 is the primary target for harness-class work.

6.5 AXIS and AXI4 flavors cover both integration styles

The AXIS flavor is a single streaming pipe; the AXI4 flavor replaces the middle with `rs_axi4_pipeline`, a memory-to-memory job chain over block RAM. Both use the same generator, checker, CSRs, and verdict logic, so a run reports itself identically whichever middle is built. Running the same deterministic campaign on both flavours is a free sanity check. The AXI4 flavor refuses a `GEN_BLOCKS` value that would exceed the job memories.

6.6 Deliberately not board-tested

The dual-decoder comparator with `ENABLE_COMPARE=1` is the sim/DV domain. The soak campaign has a default time budget and is scheduled separately. The component DV matrix exercises the codec in ways the board harness does not replicate. The erasure $f = 2t$ boundary is a known open issue, detailed in the next section.

7 Erasure Campaign

7.1 What erasure decoding adds

When `INJ_CFG.mark` is set, the injector's hit mask rides the decoder `in_erasure` sideband. The decoder is then told which symbols were corrupted, so the correction bound doubles: the decoder can correct e errors plus f erasures whenever $2e + f \leq 2t$. For RS(252,236) $t=8$, that means:

- $f = t = 8$ erasures alone should correct.
- $f = 2t = 16$ erasures alone should correct.
- $f = 2t + 1 = 17$ erasures should be refused by inspection.

7.2 Board findings

The erasure sequence was run on the Genesys 2 board:

Table 3.2: Reed-Solomon erasure boundary results.

f	Expected	Observed
8 (t)	Correct	Correct: 16/16 blocks corrected, A=B solver agreement.
16 (2t)	Correct	Flagged uncorrectable: ok/corr/unc = 0/0/16.
17 (2t+1)	Refuse by inspection	Refused: 16/16 blocks uncorrectable.

7.3 TASK-005 boundary

The $f = 2t$ case is the known open issue TASK-005 (vault/Tasks/projects/components/ecc-ip/reed-solomon/task/open/TASK-005.md): the B-stage `t_zero` bypass intends to pass $f = 16$, but the final verdict fires downstream. The $f = t$ and $f = 2t+1$ paths are correct; the $f = 2t$ path must be fixed in the component DV and then re-run on the board. This report documents the finding exactly as measured and does not fabricate a corrected result.

8 Battery Results

8.1 Per-image campaign status

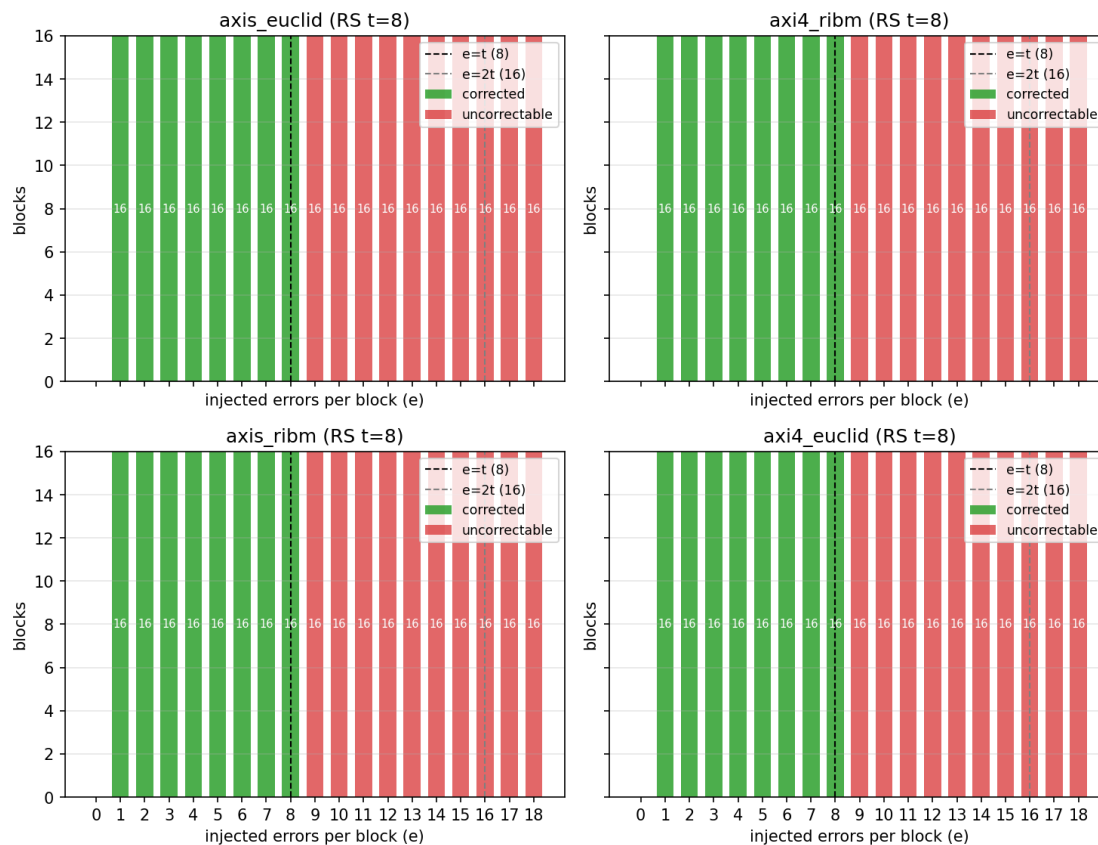
All four images passed every campaign in the 2026-10-05 battery:

Table 4.1: Reed-Solomon 2026-10-05 battery per-image status.

Image	Solver	Init	Smoke	Sweep	Random	Clusters	Localized	Badblock
axis_ribm	riBM	PASS	PASS	PASS	PASS	PASS	PASS	PASS
axis_euclid	Euclid	PASS	PASS	PASS	PASS	PASS	PASS	PASS
axi4_ribm	riBM	PASS	PASS	PASS	PASS	PASS	PASS	PASS
axi4_euclid	Euclid	PASS	PASS	PASS	PASS	PASS	PASS	PASS

8.2 Correction boundary

correction boundary: corrected vs uncorrectable by injected error count

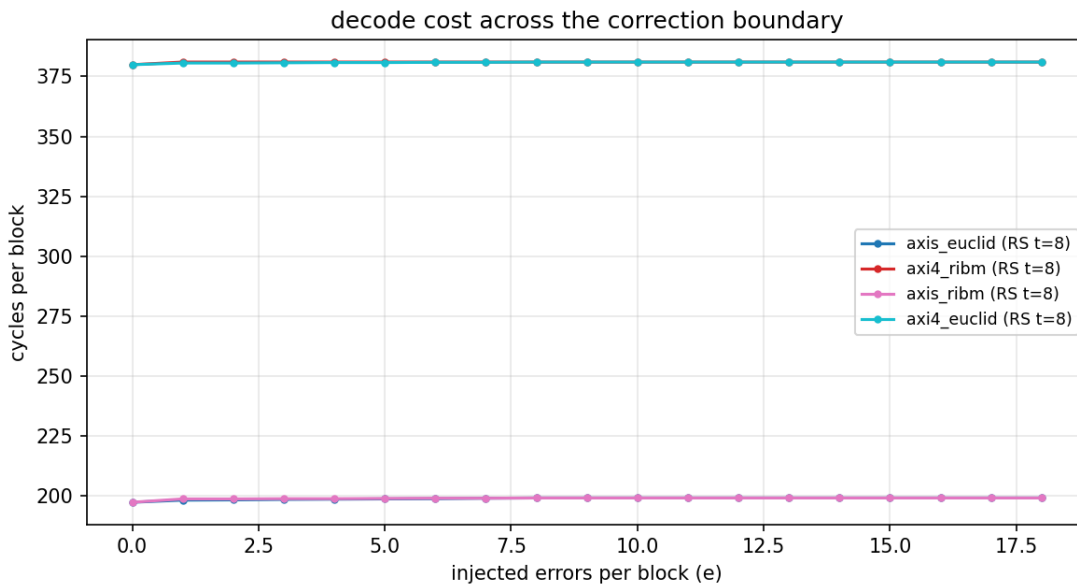


fig_1_correction_boundary.png

8.2.1 Figure 4.1: Reed-Solomon correction boundary across all four images.

8.3 Decode cost

Fig 2: decode cost



fig_2_decode_cost.png

8.3.1 Figure 4.2: Reed-Solomon decode cost across all four images.

8.4 Sweep verdict table

The sweep ran $e = 0 \dots 18$ with 16 blocks per count on every image:

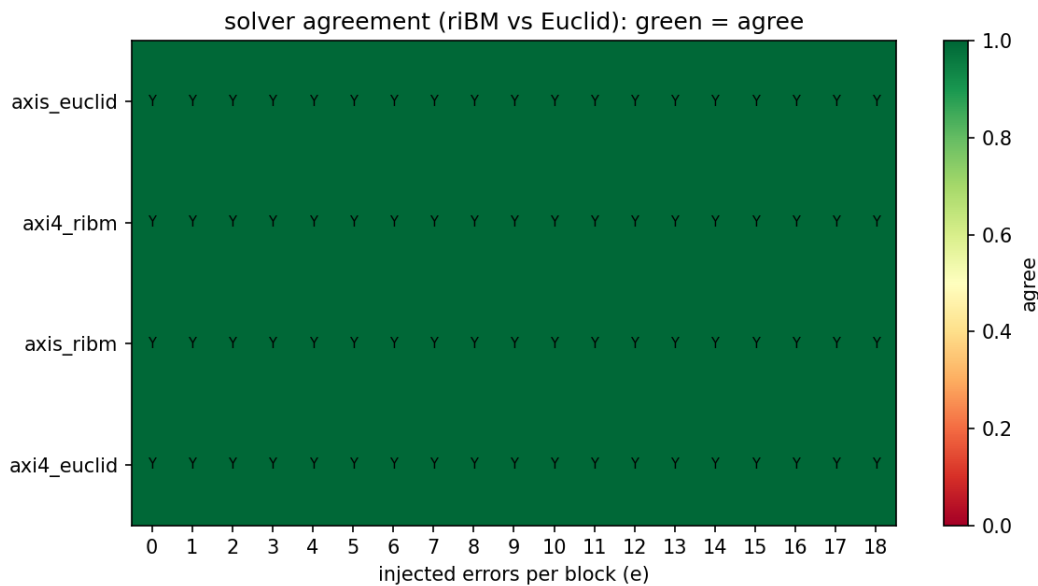
Table 4.2: Reed-Solomon sweep verdicts.

e	Verdict	Blocks	Notes
0	Correct	16	Clean decode.
1	Correct	16	16 symbols corrected.
2	Correct	16	32 symbols corrected.
3	Correct	16	48 symbols corrected.
4	Correct	16	64 symbols corrected.
5	Correct	16	80 symbols corrected.
6	Correct	16	96 symbols corrected.
7	Correct	16	112 symbols corrected.
8	Correct	16	128 symbols corrected.
9	Uncorrectable	16	Past correction boundary.
10	Uncorrectable	16	Past correction boundary.
11	Uncorrectable	16	Past correction boundary.
12	Uncorrectable	16	Past correction boundary.
13	Uncorrectable	16	Past correction boundary.
14	Uncorrectable	16	Past correction

e	Verdict	Blocks	Notes
15	Uncorrectable	16	boundary. Past correction boundary.
16	Uncorrectable	16	Past correction boundary.
17	Uncorrectable	16	Past correction boundary.
18	Uncorrectable	16	Past correction boundary.

8.5 Solver agreement

Fig 5: solver agreement



fig_5_solver_agreement.png

8.5.1 Figure 4.3: Reed-Solomon solver agreement matrix (riBM vs Euclid).

The matrix is entirely green with a Y in every cell. That means no image, at any injected error count from 0 to 18, disagreed with its sibling solver on the same deterministic workload. The absence of any red cell is the board evidence that the two key-equation solvers are consistent.

9 Soak Run Scheduled

9.1 Status at publication time

The million-block soak for Reed-Solomon was scheduled at publication time and had not yet completed. This chapter records the target and the reporting plan; it contains no fabricated numbers.

9.2 Target

- Code: RS(252,236) $t=8$, $S=4$.
- Board: Digilent Genesys 2 serial 200300B818A0.
- Target image: genesys2_axis_ribm or genesys2_axis_euclid; the AXIS flavour is used because the AXI4 job-memory cap makes long soaks impractical.
- Target volume: 1,000,000 blocks, matching the BCH soak depth.
- Host interface: UART 115200-8N1, so the effective block rate will be host-bound rather than datapath-bound.

9.3 What will be reported

When the soak completes, the following will be added to this chapter and the battery JSON artifact:

- Final block count, wall time, and blocks per second.
- Clean, corrected, and uncorrectable block counts.
- Total symbols corrected.
- Number of blocks with more than t errors and whether every one was flagged.
- Mis-decode count (must be 0 for a pass).
- The soak-timeline figure.

Until those numbers exist, the honest statement is that the run is scheduled.

10 Honest Limits

10.1 What the board evidence does not cover

- The soak is scheduled but not complete at publication time; Chapter 4 contains no soak numbers.
- The soak, once complete, will be statistical, not exhaustive. One million blocks is large, but it will not visit every possible error pattern.
- The erasure $f = 2t$ boundary is an open issue (TASK-005). $f = t$ corrects and $f = 2t+1$ refuses on the board; $f = 2t$ is flagged uncorrectable even though the design intent says it should correct.
- The dual-decoder comparator with `ENABLE_COMPARE=1` is sim-only on the board matrix. Solver agreement on the board is proven by running identical deterministic campaigns on the single-decoder images.
- BCH has no erasure path by design; that comparison is RS-only.
- The board validates only the RS(252,236) $t=8$, $S=4$ profile. Other profiles are covered in simulation and component DV.
- The component DV matrix exercises the codec in ways the board harness does not replicate.

10.2 Traceability

Every quantitative claim in this book can be traced to an artifact under stable/:

Table 5.1: Traceability from claim to artifact.

Claim	Artifact
Four images timing-clean at 100 MHz	stable/reports/genesys2_{axis_ribm,axis_euclid,axi4_ribm,axi4_euclid}/timing_summary.txt, all WNS positive, 0 failing endpoints.
7/7 campaigns PASS on all images	stable/results/2026-10-05_battery/battery.json and FINDINGS.md.
$e \leq 8$ corrected, $e > 8$ uncorrectable	stable/results/2026-10-05_battery/battery.json, sweep array.
Solver agreement matrix all green	stable/results/2026-10-05_battery/fig_5_solver_agreement.png.
$f = 2t$ erasure boundary open	vault/Tasks/projects/components/ecc-ip/reed-solomon/task/open/TASK-005.md.